

Department Overview

- **Department Name:** Department of Telecommunication Engineering
- **Faculty:** Faculty of Information & Communication Technology (FICT)
- **Accreditation:** The program is officially accredited by the **Pakistan Engineering Council (PEC)**.
- **Core Technological Focus:** Cutting-edge research and application in emerging fields including Blockchain, Artificial Intelligence (AI), AI-enabled telecommunication networks & systems, and 5G & beyond networks.
- **Industrial Specialization Disciplines:** RF & Microwave, Satellite Communication, Optical Fiber Communication, Wireless & Mobile Communication, and Analog & Digital Communication.
- **History & Impact:** Began as an undergraduate program in 2005, graduating its first batch in 2009. It equips approximately 50 students with telecommunication systems expertise each year. The department expanded to offer an MS degree program in 2013.

Program Educational Objectives (PEOs)

The program is structured to cultivate technical experts who will:

1. Investigate engineering problems using modern techniques and propose effective solutions.
2. Manage projects and become effective members of engineering teams through interpersonal skills.
3. Demonstrate high moral and ethical values, a life-long learning attitude, and fulfill societal responsibilities.

Bachelor of Science (BS) Program

Admission Requirements

- **F.Sc / Equivalent:** F.Sc (Pre-Engineering) or ICS with Mathematics and Physics from a recognized board, securing at least **60% marks**.
- **Diploma of Associate Engineer (DAE):** Completed in a relevant field with a minimum of **60% marks**.

Degree Requirements

- **Total Credit Hours:** 130 – 145
- **Total Courses:** 42 – 44
- **Minimum Passing Standing:** CGPA $\geq$ 2.0

Curriculum Plan (Program Schema by Semester)

Note: Credit hour listings follow the format (Theory + Lab).

Semester	Course Code	Course Title	Prerequisite	Credit Hours
1st	HUM-163	Functional English	None	3+0
	MATHP-118	Calculus & Analytical Geometry	None	3+0
	HUM-101	Islamic Studies / Ethical Behavior	None	2+0
	PHY-205	Applied Physics	None	3+1
	CE-162L	Computer Aided Engineering Design	None	0+1
	CS-114	Programming Fundamentals	None	2+1
	EE-103L	Electronics Workshop	None	0+1
2nd	MATHA-214	Differential Equations	Calculus & Analytical Geometry	3+0
	HUM-268	Communication Skills	None	3+0

	EE-102	Digital Logic Design	None	3+1
	EE-203	Circuit Analysis	None	3+1
	CS-212	Object Oriented Programming	Programming Fundamentals	3+1
3rd	MATHA-213	Complex Variables & Transforms	Calculus & Analytical Geometry	3+0
	HUM-102	Pakistan Studies	None	2+0
	ELE-201	Electrical Network Analysis	Circuit Analysis	2+1
	MATHP-111	Linear Algebra	None	3+0
	CS-214	Data Structures and Algorithms (IDEE-I)	—	2+1
	EE-208	Electronic Devices and Circuits	Applied Physics	3+1
4th	TE-408	Computer Communications & Networks	None	3+1

	CE-324	Microprocessors and Microcontrollers	Digital Logic Design	3+1
	ELE-202	Probability Methods in Engineering	None	3+0
	EE-261	Signals & Systems	Complex Variables and Transforms	3+0
	EE-105	Electromagnetic Theory	Calculus & Analytical Geometry	3+0
5th	TE-433	Wave Propagation & Antennas	Electromagnetic Theory	3+1
	TE-314	Communication Systems	None	3+1
	HUM-264	Technical Report Writing	None	3+0
	TE-412	Digital Signal Processing	Signals and Systems	3+1
	EE-241	Control Systems	Linear Algebra	2+1
6th	TE-442	Optical Fiber	None	3+1

		Communications		
	—	IDEE-II	—	2+1
	TE-313	Digital Communications	Probability Methods in Engineering	3+1
	MGMT-202	Entrepreneurship & Leadership	None	3+0
	TE-415	Transmission and Switching Systems	None	3+1
7th	TE-432	Microwave Engineering	Wave Propagation & Antennas	3+1
	TE-425	Emerging Wireless Technologies & RF Planning	None	3+0
	TE-422	MBC I (Telecom Traffic Engineering)	Probability Methods in Engineering	3+0
	ECON-105	Engineering Economics	None	3+0

	TE-452L	Final Year Project-I	—	0+3
8th	MGMT-304	Organizational Behavior or Sociology	None	3+0
	TE-423	MBC II (Telecom Policies & Regulation)	—	3+0
	TE-441 / TE- 409	MBC III (Satellite Comm. or Network Security)	—	3+0
	HUM-302	Professional Practices	None	3+0
	TE-453L	Final Year Project-II	—	0+3